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#include <WiFi.h>

#include <WiFiClientSecure.h>

#include <PubSubClient.h>

#include <DHT.h>

#include <LiquidCrystal_I2C.h>


// ===== WIFI =====

const char* ssid = "Tes";

const char* password = "123abcde";


// ===== MQTT =====

const char* mqtt_server =
"699e728e48d34ed1a9bd5e7fc8f2d063.s1.eu.hivemq.cloud";

const int mqtt_port = 8883;

const char* mqtt_user = "esp32_client";

const char* mqtt_password = "Suhu2026!@#";


// ===== MQTT TOPIC =====

const char* topic_control = "iot/control";

const char* temp_topic = "iot/suhu/temperature";

const char* hum_topic = "iot/suhu/humidity";

const char* soil_status_topic = "iot/soil/status";

const char* status_topic = "iot/status";


// ===== MQTT =====

WiFiClientSecure client;

PubSubClient mqttClient(client);

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// ===== PIN =====

#define RELAY_HEATER 25

#define RELAY_UV 26

#define RELAY_PUMP 27

#define RELAY_EXHAUST 14

#define LED_MERAH 23

#define LED_HIJAU 19

#define SOIL_PIN 34

#define DHTPIN 4

#define DHTTYPE DHT22


// ===== OBJECT =====

DHT dht(DHTPIN, DHTTYPE);

LiquidCrystal_I2C lcd(0x27, 16, 2);


// ===== VAR =====

bool running = false;

unsigned long startTime = 0;


// ===== CALLBACK MQTT =====

void callback(char* topic, byte* payload, unsigned int length) {

    String message = "";

    for (int i = 0; i < length; i++) {

        message += (char)payload[i];

    }

    message.trim();

    Serial.print("MQTT MESSAGE: ");

    Serial.println(message);

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// ===== START =====

if (message == "start") {

    running = true;

    startTime = millis(); // Catat waktu awal mulai


    // Saat start dimulai, Pompa langsung ON (Relay Active Low = LOW)
    digitalWrite(RELAY_PUMP, LOW);
    digitalWrite(RELAY_HEATER, HIGH);
    digitalWrite(RELAY_UV, HIGH);
    digitalWrite(RELAY_EXHAUST, HIGH);
    mqttClient.publish(status_topic, "RUNNING");
}


// ===== STOP =====

if (message == "stop") {

    stopSystem();

}

}


// ===== CONNECT MQTT =====

void reconnectMQTT() {

    while (!mqttClient.connected()) {

        Serial.println("Connecting MQTT...");

        String clientId = "ESP32-" + String(random(0xffff), HEX);

        if (mqttClient.connect(clientId.c_str(), mqtt_user, mqtt_password)) {

            Serial.println("MQTT CONNECTED");

            mqttClient.subscribe(topic_control);

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    } else {
        Serial.print("FAILED MQTT: ");
        Serial.println(mqttClient.state());
        delay(3000);
    }
}

// ===== STOP SYSTEM =====

void stopSystem() {
    running = false;
    digitalWrite(RELAY_HEATER, HIGH);
    digitalWrite(RELAY_UV, HIGH);
    digitalWrite(RELAY_PUMP, HIGH);
    digitalWrite(RELAY_EXHAUST, HIGH);
    digitalWrite(LED_MERAH, HIGH);
    digitalWrite(LED_HIJAU, LOW);
    mqttClient.publish(status_topic, "SELESAI");
}

// ===== SETUP =====

void setup() {
    Serial.begin(115200);

    // ===== PINMODE =====

    pinMode(RELAY_HEATER, OUTPUT);
    pinMode(RELAY_UV, OUTPUT);
    pinMode(RELAY_PUMP, OUTPUT);

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pinMode(RELAY_EXHAUST, OUTPUT);

pinMode(LED_MERAH, OUTPUT);

pinMode(LED_HIJAU, OUTPUT);


// ===== DEFAULT OFF =====

digitalWrite(RELAY_HEATER, HIGH);

digitalWrite(RELAY_UV, HIGH);

digitalWrite(RELAY_PUMP, HIGH);

digitalWrite(RELAY_EXHAUST, HIGH);

digitalWrite(LED_MERAH, HIGH);

digitalWrite(LED_HIJAU, LOW);


// ===== LCD =====

lcd.init();

lcd.backlight();

lcd.print("Connecting WiFi");


// ===== DHT =====

dht.begin();


// ===== WIFI =====

WiFi.begin(ssid, password);

client.setInsecure();

while (WiFi.status() != WL_CONNECTED) {

    delay(500);

    Serial.print(".");

}

Serial.println("");

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Serial.println("WIFI CONNECTED");

lcd.clear();

lcd.print("WiFi Connected");


// ===== MQTT =====

mqttClient.setServer(mqtt_server, mqtt_port);

mqttClient.setCallback(callback);
}


// ===== LOOP =====

void loop() {

  if (!mqttClient.connected()) {

    reconnectMQTT();

  }

  mqttClient.loop();


// ===== BACA SOIL =====

int soilRaw = 0;

for (int i = 0; i < 5; i++) {

  soilRaw += analogRead(SOIL_PIN);

  delay(10);

}

soilRaw /= 5;


// ===== KALIBRASI SENSOR =====

int soilDry = 3200; // nilai saat kering

int soilWet = 1400; // nilai saat basah

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// Konversi ke persen

int soil = map(soilRaw, soilDry, soilWet, 0, 100);
soil = constrain(soil, 0, 100);


// ===== BACA SUHU =====

float suhu = dht.readTemperature();

if (isnan(suhu)) {
    return;
}


// ===== PUBLISH SELALU =====

char tempStr[8];
char humStr[8];
dtostrf(suhu, 1, 2, tempStr);
dtostrf(soil, 1, 2, humStr);
mqttClient.publish(temp_topic, tempStr);
mqttClient.publish(hum_topic, humStr);


// ===== STATUS SEPATU =====

if (soil <= 10) {
    mqttClient.publish(soil_status_topic, "KERING");
} else if (soil <= 40) {
    mqttClient.publish(soil_status_topic, "LEMBAB");
} else {
    mqttClient.publish(soil_status_topic, "BASAHA");
}


// ===== RUNNING =====

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if (running) {

    mqttClient.publish(status_topic, "RUNNING");

    digitalWrite(LED_MERAH, LOW);

    digitalWrite(LED_HIJAU, HIGH);


    lcd.setCursor(0, 0);

    lcd.print("Running...  ");

    lcd.setCursor(0, 1);

    lcd.print("T:");

    lcd.print(suhu);

    lcd.print(" H:");

    lcd.print(soil);

    lcd.print("% ");


    // Hitung durasi waktu sejak tombol start ditekan
    unsigned long timeElapsed = millis() - startTime;


    // FASE 1: Detik ke-0 sampai ke-5 -> Pompa Hidup
    if (timeElapsed < 5000) {

        digitalWrite(RELAY_PUMP, LOW);    // Pompa ON

        digitalWrite(RELAY_HEATER, HIGH); // Heater OFF

        digitalWrite(RELAY_UV, HIGH);     // UV OFF

        digitalWrite(RELAY_EXHAUST, HIGH); // Kipas OFF

    }

    // FASE 2: Detik ke-5 sampai ke-10 -> Pompa Mati & Jeda Tunggu (Delay 5 Detik)
    else if (timeElapsed >= 5000 && timeElapsed < 10000) {

        digitalWrite(RELAY_PUMP, HIGH);    // Pompa OFF

        digitalWrite(RELAY_HEATER, HIGH); // Heater OFF
    }
}

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digitalWrite(RELAY_UV, HIGH);    // UV OFF

digitalWrite(RELAY_EXHAUST, HIGH); // Kipas OFF
}

// FASE 3: Detik ke-10 ke atas -> Heater, UV, & Exhaust Aktif Sesuai Aturan Suhu

else {

    digitalWrite(RELAY_PUMP, HIGH);    // Pastikan pompa tetap OFF


    // ===== LOGIKA HEATER & EXHAUST =====

    // Heater hidup di bawah 60 C, Exhaust hidup di atas atau sama dengan 60 C (ketika
heater mati)

    if (suhu < 60.0) {

        digitalWrite(RELAY_HEATER, LOW); // Heater ON

        digitalWrite(RELAY_EXHAUST, HIGH); // Exhaust OFF

    } else {

        digitalWrite(RELAY_HEATER, HIGH); // Heater OFF

        digitalWrite(RELAY_EXHAUST, LOW); // Exhaust ON (Overheat Protection)

    }


    // ===== LOGIKA UV =====

    // UV hidup bersama heater, hanya jika suhu di bawah 40 C

    if (suhu < 40.0) {

        digitalWrite(RELAY_UV, LOW);    // UV ON

    } else {

        digitalWrite(RELAY_UV, HIGH);    // UV OFF

    }

}

// ===== AUTO STOP =====

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if (soil <= 10) {  
    mqttClient.publish(status_topic, "SELESAI");  
    stopSystem();  
}  
}  
  
// ===== STANDBY =====  
  
else {  
    mqttClient.publish(status_topic, "OFF");  
    digitalWrite(LED_MERAH, HIGH);  
    digitalWrite(LED_HIJAU, LOW);  
  
    lcd.setCursor(0, 0);  
    lcd.print("STANDBY   ");  
    lcd.setCursor(0, 1);  
    lcd.print("T:");  
    lcd.print(suhu);  
    lcd.print(" H:");  
    lcd.print(soil);  
    lcd.print("% ");  
}  
  
delay(1000);  
}
```